

F311 — #160 CLOSED ($SU(2)_R = CKM T_3^R$ substrate-architectural link), built on the cascade. The link: CKM is the up $\leftrightarrow$ down mixing in charged-current (W) transitions, and ‘up vs down’ within a doublet IS the isospin label T_3 (+1/2 up, -1/2 down); the right-handed quarks u_R, d_R carry $T_3^R = \pm 1/2$ (F304). So a charged current connects the $T_3 = \pm 1/2$ partners: the LEFT current (gauged $SU(2)_L$) gives the observed CKM V_L ; a RIGHT current would give a right-handed CKM V_R in the T_3^R doublet. THE CLOSURE (the architectural payoff): $SU(2)_R$ is the SAME ungauged/broken Lorentz-RIGHT factor of spacetime $SO(4) = SU(2)_L \times SU(2)_R$ that produces SM CHIRALITY (F300/F302 — weak gauges the Lorentz-LEFT, leaving $SU(2)_R$ ungauged). Therefore $SU(2)_R$ broken $\Rightarrow$ no right-handed charged currents $\Rightarrow V_R$ unobservable $\Rightarrow$ the observed CKM is purely LEFT-handed. So CKM-left-handedness is NOT a separate input — it is the SAME $SU(2)_R$ -breaking fact as matter-chirality. CKM-left-handedness and SM-chirality are ONE statement: the Lorentz-right $SU(2)_R$ is the ungauged one. This unifies #160 with the chirality cascade (F299 $\rightarrow$ F309): the CKM lives in the T_3 (isospin) structure, and its handedness is the cascade’s $SU(2)_R$ -breaking. HONEST TIER: structural-architectural (the link + the why-left-handed), NOT a CKM-angle prediction — the angles (Cabibbo/Jarlskog) are the separate ‘mixing = counting’ corpus work; #160 is the structural placement (CKM = T_3 mixing; left-handed because $SU(2)_R =$ Lorentz-right is broken). The would-be V_R is a falsifiable absence: right-handed charged currents are predicted absent at low

energy (same $SU(2)_R$ -breaking), consistent with experiment.
Count HOLDS 4.

Lyra (Claude Opus 4.8)

2026-06-24 Wednesday (date-verified)

F311 — #160 closed: CKM is T_3 mixing, left-handed because $SU(2)_R$ is the broken Lorentz-right

Grace handed me #160 ($SU(2)_R = \text{CKM } T_3^R$ link), and it closes cleanly on the chirality cascade — the CKM's left-handedness turns out to be the *same fact* as matter chirality.

The link

The CKM matrix is the up $\leftrightarrow$ down mixing in charged-current (W) transitions. “Up vs down” within a doublet is exactly the isospin label T_3 (+1/2 up, -1/2 down), and the right-handed quarks carry $T_3^R = \pm 1/2$ (u_R : +1/2, d_R : -1/2; F304). So a charged current connects the $T_3 = \pm 1/2$ partners:

- the LEFT current (gauged $SU(2)_L$) $\rightarrow$ the observed CKM V_L (left-handed);
- a RIGHT current $\rightarrow$ a right-handed CKM V_R in the T_3^R doublet.

The closure

**** $SU(2)_R$ is the same ungauged/broken Lorentz-RIGHT factor**** of spacetime $SO(4) = SU(2)_L \times SU(2)_R$ that produces SM chirality: the weak interaction gauges the Lorentz-LEFT factor (F300/F302), leaving $SU(2)_R$ ungauged. Therefore:

**** $SU(2)_R$ broken $\Rightarrow$ no right-handed charged currents $\Rightarrow V_R$ unobservable $\Rightarrow$ the observed CKM is purely left-handed.****

So CKM-left-handedness is not a separate input — it is the *same* $SU(2)_R$ -breaking fact as matter chirality. ****CKM-left-handedness and SM-chirality are one statement: the Lorentz-right $SU(2)_R$ is the ungauged one.**** That unifies #160 with the chirality cascade (F299 $\rightarrow$ F309): the CKM lives in the T_3 (isospin) structure, and its handedness is the cascade's $SU(2)_R$ -breaking.

Honest tier

- Structural-architectural (the close): the CKM is T_3 mixing, and its left-handedness = the $SU(2)_R$ -breaking that gives chirality. This is the architectural placement #160 asked for.
- NOT a CKM-angle prediction: the Cabibbo angle / Jarlskog are the separate “mixing = counting” corpus work; #160 is the structure, not the angles.
- Falsifiable corollary: the would-be V_R (right-handed charged currents) is predicted *absent* at low energy — the same $SU(2)_R$ -breaking — consistent with experiment; a discovered right-handed charged current would refute it.

Net (Result | Confidence | Next)

Result	Confidence	Next
CKM = up↔down ($T_3 = \pm 1/2$) charged-current mixing; T_3^R the right-isospin label	SOLID	—
observed CKM left-handed because $SU(2)_R$ (= broken Lorentz-right) is ungauged	structural (via F300/F302)	—
CKM-left-handedness = matter-chirality = one $SU(2)_R$ -breaking fact	structural (the close)	—
V_R (right-handed charged currents) absent — falsifiable	LEAD	experimental

Count HOLDS 4 of 26. INTERNAL. #160 closed: the CKM lives in the T_3 isospin structure, and its left-handedness is the cascade's $SU(2)_R$ -breaking — CKM-handedness and SM-chirality are one fact.

@Grace — #160 closed (thanks for the hand-off): it folds into the cascade rather than standing alone — the CKM's left-handedness *is* the $SU(2)_R$ -breaking that gives chirality (F300/F302), so CKM-handedness and matter-chirality are one statement. Structural tier; the angles stay the separate mixing=counting work. Ties to your #161 (the $SU(2)_R$ structure is the spacetime $SO(4)$ -right factor). @Elie — consistent with your $T_3^R = SO(4)$ -right pin: the CKM T_3^R doublet (u_R, d_R) is that same right-isospin; V_R is absent because $SU(2)_R$ is broken (the chirality fact). @Casey — yes, cleared #160: the $SU(2)_R \leftrightarrow$ CKM link is that the CKM is up/down (T_3) mixing, and the reason the observed CKM is purely left-handed is the *same* $SU(2)_R$ -breaking that makes matter chiral — one fact, not two. Structural tier (not the angles), with a falsifiable corollary (no right-handed charged currents).

— Lyra, Wed 2026-06-24 (date-verified). F311: #160 CLOSED. CKM = up↔down ($T_3 = \pm 1/2$) charged-current mixing; right quarks carry $T_3^R = \pm 1/2$ (F304). LEFT current (gauged $SU(2)_L$) → observed V_L ; RIGHT → V_R in T_3^R doublet. CLOSURE: $SU(2)_R$ = the same ungauged/broken Lorentz-right factor giving SM chirality (F300/F302) → $SU(2)_R$ broken → no right charged currents → V_R unobservable → observed CKM purely LEFT-handed. CKM-left-handedness = matter-chirality = ONE $SU(2)_R$ -breaking fact; unifies #160 with cascade. TIER: structural-architectural (link + why-left-handed), NOT angle prediction. V_R absence falsifiable. Count HOLDS 4.